No. of Printed Pages: 05
SESSION: 2025-26

SUPPLEMENTARY PRE BOARD EXAMINATION
CLASS X
MATHEMATICS (041)

Time: 3 Hours

Maximum Marks : 80

Admn. No.	Name of Student	Signature of Invigilator with date
14168	Vyoma Jagya.	

General Instructions:
1. All questions are compulsory
2. Section A contains 20 Multiple choice Questions (MCQs) of 1 mark each.
3. Section B contains 5 questions of 2 marks each.
4. Section C contains 6 questions of 3 marks each.
5. Section D contains 4 questions of 5 marks each.
6. Section E contains 3 Case Study based questions of 4 marks each.

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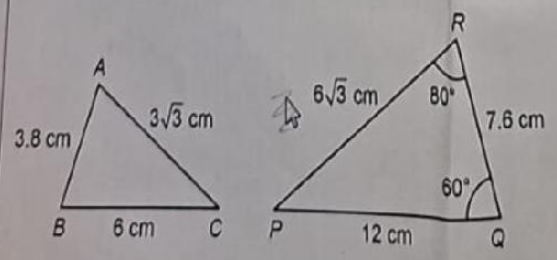
SECTION A (1x20=20)

Q.No.	Questions	Marks
1.	Which of the following cannot be the unit digit of 8^n, where n is a natural number? (a) 4 (b) 2 (c) 0 (d) 6	1
2.	Let $x = a^2b^3c^n$ and $y = a^3b^mc^2$, where a, b and c are prime numbers. If LCM of x and y is $a^3b^4c^3$, then value of m + n is : (a) 10 (b) 5 (c) 6 (d) 7	1
3.	If the zeroes of the polynomial $abx^2 + b^2x + 2a$ are reciprocal of each other, then the value of b is: (a) 2 (b) -2 (c) $\frac{1}{2}$ (d) $\frac{-1}{2}$	1
4.	The value of k for which the system of equations: 3x- ky=7 and 6x +10y=3 is inconsistent is: (a) -10 (b) 5 (c) -5 (d) 7	1
5.	Which of the following quadratic equation has real and equal roots? (a) $(x+1)^2 = 2x+1$ (b) $x^2 + x = 0$ (c) $x^2 - 4 = 0$ (d) $x^2 + x + 1 = 0$	1
6.	Points P, Q and R divide the line segment AB in four equal parts. The point Q divides PB in the ratio: (a) 1:3 (b) 2:3 (c) 1:2 (d) 1:1	1
7.	What is the distance of the point (4, 2) from the origin? (a) $\sqrt{20}$ (b) 4 (c) 2 (d) 20	1
8.	ΔABC and ΔPQR are shown in the given figures. The measure of $\angle C$ is:  (a) 140° (b) 80° (c) 60° (d) 40°	1

9.	tan2A = 3tanA is true, when measure of ∠A is: (a) 90° (b) 60° (c) 45° (d) 30°	1
10.	$\frac{1-\tan^2 30°}{1+\tan^2 30°}$ is equal to: (a) Sin 60° (b) tan 60° (c) cos 60° (d) sec 60°	1
11.	The volumes of two spheres are in the ratio 64:27. Find the ratio of their surface areas. (a) 9:16 (b) 4:9 (c) 9:4 (d) 16:9	1
12.	The area of a sector of a circle of radius 5cm formed by an arc of length 3.5cm is: (a) 7.5 cm² (b) 7.75 cm² (c) 8.5 cm² (d) 8.75 cm²	1
13.	The area of a quadrant of a circle whose circumference is 22cm is: (a) 9.25 cm² (b) 7.15 cm² (c) 3.65 cm² (d) 9.63 cm²	1
14.	If two tangents inclined at an angle 60° are drawn to a circle of radius 3cm, then length of each tangent is equal to: (a) $\frac{3}{2}\sqrt{3}$ cm (b) 6 cm (c) 3 cm (d) $3\sqrt{3}$ cm	1
15.	A card is drawn from a deck of 52 cards. The event E is that card is not an ace of hearts. The number of outcomes favorable to E is: (a) 4 (b) 13 (c) 48 (d) 51	1
16.	One ticket is drawn at random from a bag containing tickets numbered 1 to 40. The probability that the selected ticket has a number which is not a multiple of 5 is: (a) $\frac{1}{5}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$ (d) $\frac{1}{3}$	1
17.	Consider the data: **Classes**: 65-85, 85-105, 105-125, 125-145, 145-165 **Frequency**: 4, 5, 13, 20, 14 The modal class is: (a) 105-125 (b) 125-145 (c) 145-165 (d) None of these	1
18.	From a point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. The radius of the circle is: (a) 7 cm (b) 12 cm (c) 15 cm (d) 24.5 cm	1
19.	**Assertion (A):** For any two prime numbers p and q, their HCF is 1 and LCM is p+ q. **Reason (R):** For any two natural numbers, HCF x LCM = product of numbers. (a) Both (A) and (R) are true, and (R) is the correct explanation of (A). (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (c) (A) is true but (R) is false. (d) (A) is false but (R) is true	1
20.	**Assertion (A):** The value of sin $\theta = \frac{4}{3}$ is not possible. **Reason (R):** Hypotenuse is the largest side in any right-angled triangle (a) Both (A) and (R) are true, and (R) is the correct explanation of (A). (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (c) (A) is true but (R) is false. (d) (A) is false but (R) is true	1

SECTION B

21.	If the sum of first 14 terms of an AP is 1050 and its first term is 10, find the 20th term. **OR** Which term of the AP: 3, 15, 27, 39, ... will be 132 more than its 54th term?	2
22.	Quadrilateral ABCD circumscribes a circle. If AB= 6cm, BC= 7cm and CD=4cm, then find AD.	2
23.	The length of the minute hand of a clock is 10 cm. Find the area swept by hand in 35 minutes. **OR** 3 animals are tied at each corner of a triangular park with a rope of length 7m each. Find the area grazed by the 3 animals.	2
24.	Evaluate: $2(\cos^2 45° - \tan^2 60°) + 6(\sin^2 45° + \tan^2 30°)$	2
25.	S and T are points on sides PR and QR of ΔPQR such that $\angle P = \angle RTS$. Show that $\Delta RPQ \sim \Delta RTS$.	2

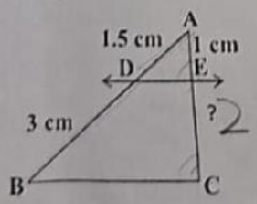
(3x6=18)

SECTION C

26.	If α and β are zeroes of polynomial $p(x)= x^2-3x-10$, then find the value of $\alpha^2+\beta^2-\alpha\beta$	3
27.	Prove that $3+2\sqrt{5}$ is an irrational number.	3
28.	Two dice are thrown simultaneously. Find the probability: (a) Sum an even no. (b) Product of nos. less than 9 (c) a doublet	3
29.	If $\cos\theta + \sin\theta = \sqrt{3}$, prove that $\tan\theta + \cot\theta = 1$. **OR** Prove the following identity: $\sec A (1 - \sin A)(\sec A + \tan A) = 1$	3
30.	Solve: $37x + 63y = 137$ $63x + 37y = 163$ **OR** The sum of the digits of a two-digit number is 9. Also, nine times this number is twice the number obtained by reversing the order of the digits. Find the number.	3
31.	Two concentric circles are of radii 10cm and 6cm. Find the length of the chord of larger circle which touches the smaller circle.	3

(5x4=20)

SECTION D

32.	Prove that, if a line is drawn parallel to one side of a triangle to intersect other two sides in distinct point, then the other two sides are divided in the same ratio. Using the above theorem, find the length of EC if DE ∥ BC.  	5

$x = 3 + 2y$ | $90 = 3 + 2y$

33.	A cottage industry produces a certain number of pottery articles in a day. It was observed on a particular day that the cost of production of each article (in rupees) was 3 more than twice the number of articles produced on that day. If the total cost of production on that day was ₹ 90, find the number of articles produced and the cost of each article. $6, 3y = 90$	5
34.	A tent is in the shape of a cylinder surmounted by a conical top. If the height and diameter of the cylindrical part are 2.1 m and 4 m respectively, and the slant height of the top is 2.8 m, find the area of the canvas used for making the tent. Also, find the cost of the canvas of the tent at the rate of ₹ 500 per m^2. **OR** A solid consisting of a right circular cone of height 120 cm and radius 60 cm standing on a hemisphere of radius 60 cm is placed upright in a right circular cylinder full of water such that it touches the bottom. Find the volume of water left in the cylinder, if the radius of the cylinder is 60 cm and its height is 180 cm	5
35.	Find the missing frequency(f) if the mean of the following data is 52.	5

Classes	Frequency	x_i	$f_i x_i$
10-20	5	15	75
20-30	3	25	75
30-40	4	35	140
40-50	f	45	45f
50-60	2	55	110
60-70	6	65	390
70-80	13	75	975
	231		

$52 = \frac{f_i x_i}{f_i}$

OR

Find the median:

Classes	Frequency
0-10	5
10-20	4
20-30	8
30-40	12
40-50	16
50-60	15
60-70	10
70-80	8
80-90	5
90-100	2

$\text{Median} = L + \left(\frac{\frac{n}{2} - C.f}{f}\right) \times h$

$x = 3 + 2y$

$90 = 90(3+2y)$

$90 = 90(3+2y)$

$90 = 270 + 180y$

$-180 = +180y$

$-1 = y$

$x = 3 - 2$

$x = 1$

$x = 3 + 2y - (i)$

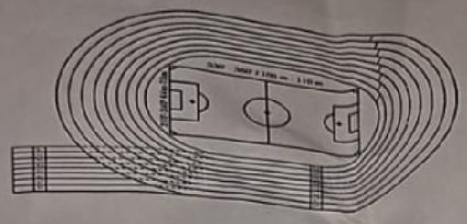
$90 =$

$75 + 75 + 150 + 140 + 290 + 45 + 335 + 110 + 445 + 390 + 835 + 975 + 1810$

SECTION E

(4x3=12)

36. In order to organise, Annual Sports Day, a school prepared an eight lane running track with an integrated football field inside the track area as shown below:



The length of innermost lane of the track is 400 m and each subsequent lane is 7.6 m longer than the preceding lane.

Based on given information, answer the following questions, using concept of Arithmetic Progression.

(i) What is the length of the 6th lane? 1

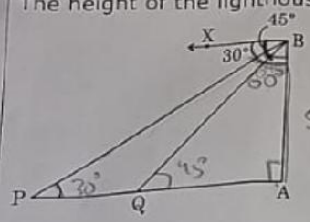
(ii) How long is the 8th lane than that of 4th lane? 1

(iii) While practicing for a race, a student took one round each in first six lanes. Find the total distance covered by the student. 2

OR

A student took one round each in lane 4 to lane 8. Find the total distance covered by the student. 2

37. A lighthouse stands tall on a cliff by the sea, watching over ships that pass by. One day a ship is seen approaching the shore and from the top of the lighthouse, the angles of depression of the ship are observed to be 30° and 45° as it moves from point P to point Q. The height of the lighthouse is 50 metres.



Based on the information given above, answer the following questions:

(i) Find the measures of ∠BPA and ∠BQA.

(ii) Find the distance of the ship from the base of the lighthouse when it is at point Q, where the angle of depression is 45°.

(iii) Find the distance travelled by the ship between points P and Q.

OR

If the ship continues moving towards the shore and takes 10 minutes to travel from Q to A, calculate the speed of the ship in km/h, from Q to A.

38. The city planning department is designing a new park in a rectangular land ABCD. The coordinates of the vertices of the land are A(2, 3), B(2,7), C(8, 7) and D(8, 3). Based of the given information, answer the following questions:

(i) Find the length and width of the park. 1

(ii) Determine the coordinates of the mid-point of diagonal AC. 1

(iii) Verify if the diagonals of the rectangle are equal. 2

OR

Find the coordinates of the point which divides AB in the ratio 2:3 2

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